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SURVEILLANCE CAMERA Using RaspiCam And ANDROID APP

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This project describes how to build a surveillance camera based on Raspberry Pi (Raspi) that records HD video only when something moves in the monitored area. Live feed can be viewed from any Web browser, including one on a mobile phone.

Security cameras are common in most industries around the world. Their applications can range from preventing theft or vandalism to traffic and weather monitoring and much more. Due to a small form factor, affordable price and low power consumption, Raspberry Pi can be easily integrated in a surveillance camera.

In this project we will first connect Raspi camera to Raspi, and stream live video from it. We can also watch the streaming of content from our Android device. We will also take photographs with different effects.

Raspi is a credit-card-sized low-cost microcomputer that can run Linux operating system, and has endless extension possibilities.

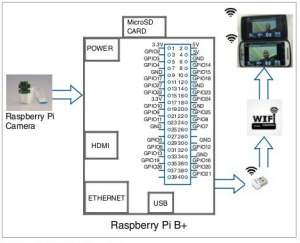


Fig. 1: Raspi and RaspiCam

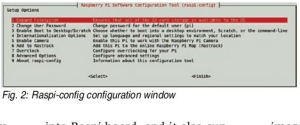


Fig. 2: Raspi-config configuration window

Hardware components

We require the following hardware for this project:

Raspberry Pi model B+. This model of Raspi system runs at 700MHz and its 512MB RAM supports HD video.

Raspberry Pi camera module. This module is specially built for the Raspi microcomputer. It has a connector that can be plugged directly into Raspi board, and it also supports HD video.

Power supply. Any power supply can be used that has a micro-USB plug, and which can supply at least 1A of current.

MicroSD card. Raspi does not have any storage on board, so you need to use a MicroSD card to install and run the operating system for this device.

Wi-Fi USB adaptor. You need this to connect the camera to your network.

Block diagram of the project is shown in Fig. 1.

Software

Get the software ready. To start, you need Raspi with an Internet connection. Update its operating system to ensure that it has all the latest features and drivers. Run the command given below, in the terminal, to update the operating system installed on Raspi:

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

Install RaspiCam Remote for Android. RaspiCam Remote is an app to view Raspi camera module on your Android device. In this project:

1. No software needs to be installed or configured on Raspi; it uses standard NOOBS setup.

2. Pictures can be saved to your phone's gallery.

3. Continuous video mode (around one frame per second) for camera monitoring can be used on your Android device.

4. It is easily configurable and allows testing the camera's image filters.

Configure raspi-config. Raspi-config is Raspi's configuration tool, written and maintained by Raspbian operating system.

You will be shown raspi-config on booting into Raspbian for the first time. To open the configuration tool, run the following from the command line:

```
$ sudo raspi-config
```

You will see a blue screen with options in a gray box in the centre as shown in Fig. 2.

Use up and down arrow keys to move the highlighted selection between the options available. In order to use Raspi camera module, select Enable Camera option as shown in Fig. 2. This will ensure that at least 128MB of RAM is dedi-